

# Operationalising the Three-Axes Framework: A Composite-Indicator Companion to *AI-Proliferation and Middle Powers*

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Companion to the working paper *AI-Proliferation and Middle Powers: Preparation and Response Mechanisms* (2026)

June 12, 2026

## Abstract

The working paper *AI-Proliferation and Middle Powers* argues that fourteen named middle powers differ along three axes—*capacity depth*, *governance orientation*, and *infrastructure posture*—and that the tractability of preparedness actions depends on a country’s position along them, yet it never empirically positions any country. This companion study closes that gap with the Middle-Power AI-Proliferation Preparedness Index (M-PAPI): a reproducible composite indicator built from fifteen publicly sourced indicators following the OECD/JRC *Handbook on Constructing Composite Indicators*. We report a literature-weighted ranking led by the United Kingdom (2.314) and South Korea (2.185) and trailed by India (0.106) and Taiwan (0.057); the top-five set {UK, Korea, France, EU, Japan} is identical across equal-, PCA-, and literature-weighting schemes. Six hypotheses are pre-registered with numeric pass criteria and resolved mechanically: two are fully supported (cluster heterogeneity; insensitivity of the top tier to any single analyst-coded indicator) and four are partially supported with specific, named caveats (cross-axis separability, axis dimensionality, ranking robustness, and tier-membership stability). A vulnerability overlay maps axis scores onto the paper’s three attack vectors, and counterfactual scenarios quantify per-country policy leverage. The ranking is strongly rank-concordant with the independent Oxford Insights Government AI Readiness Index 2024 (Spearman  $\rho = 0.857$  over the thirteen jurisdictions both cover), evidence of external concurrent validity. All outputs regenerate byte-for-byte under a pinned seed and an offline data snapshot. The index empirically grounds the paper’s §1–§2 premise—the middle powers are differentiable, and the most-exposed tier is identifiable—while explicitly declining the §3–§4 claims it cannot yet test.

**Keywords:** composite indicators; AI governance; middle powers; AI proliferation; reproducible research; sensitivity analysis.

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## D Reproducibility

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### 1 Introduction

The development of frontier artificial intelligence is concentrated in a small number of laboratories in the United States and China, leaving *middle powers*—states with meaningful but non-dominant international influence—largely in the position of responding to, rather than shaping, the technology’s trajectory. The working paper *AI-Proliferation and Middle Powers: Preparation and Response Mechanisms* (Teague et al., 2026) argues that, as access to highly capable model weights diffuses, middle powers face a structurally under-addressed vulnerability to three proliferation-enabled attack vectors—cyber-attacks, chemical/biological/radiological/nuclear (CBRN) threats, and influence operations—and that national-level preparedness is a more tractable priority than contesting the frontier. To organise that argument the paper distils two widely cited “building blocks of AI power” frameworks into three axes along which middle powers differ: *capacity depth*, *governance orientation*, and *infrastructure posture*.

The paper names fourteen middle powers—the United Kingdom, Canada, France, Germany, Japan, India, Israel, Singapore, South Korea, Sweden, Saudi Arabia, the United Arab Emirates, the European Union, and Taiwan (Wang et al., 2024)—and repeatedly invokes “tractability depending on a country’s position within the three axes,” but it never *positions* any country on those axes. This companion study supplies that missing empirical layer.

**Contribution.** We construct the Middle-Power AI-Proliferation Preparedness Index (M-PAPI), a transparent composite indicator that places each of the fourteen middle powers on the three axes using fifteen publicly available indicators, and we report per-country rankings, a per-vector vulnerability overlay, and counterfactual policy scenarios. The study is deliberately *not* new theory; it is empirical scaffolding that makes the working paper’s per-country claims auditable, replicable, and updatable. We follow the OECD/JRC *Handbook on Constructing Composite Indicators* (OECD and European Commission, Joint Research Centre, 2008) throughout and report three weighting schemes side by side with a full sensitivity battery.

**Scope.** M-PAPI operationalises the paper’s §1 (axes and country set) and §2 (attack vectors) only. The paper’s §3 (trigger-event ladder) and §4 (detection/escalation/mitigation checklist) are *not* operationalised: a country-year, AI-attributed incident dataset would be required for §3, and the §4 actions appear here only as the counterfactual policy scenarios of Section 5, not as country-year preparedness outcomes. The index measures *relative* position within the fourteen-country reference frame; it does not claim that a high score means a country is “safe.”

**Pre-registration.** Six falsifiable hypotheses (H1–H6) are stated with numeric pass criteria before estimation (Table 6) and resolved mechanically against those criteria. Reporting the partial-support outcomes explicitly—rather than only the favourable ones—is a deliberate feature of the design.

**Table 1:** The three axes of the working paper’s framework and the M-PAPI indicators assigned to each. Indicator construction and sources are detailed in Section 3.

| Axis                   | Definition (working paper §1)                                          | Indicators                                                                                                                  |
|------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Capacity Depth         | Domestic technical talent, AISI-equivalent institutions, AI R&D output | C1 notable models; C2 training compute; C3 AI publications; C4 AI patents; C5 AISI presence                                 |
| Governance Orientation | AI-governance maturity, alliance posture, bilateral lab agreements     | G1 national AI strategy; G2 ITU GCI; G3 V-Dem liberal democracy; G4 lab-MoU count; G5 IGSC member firms; G6 Australia Group |
| Infrastructure Posture | Domestic compute, ICT infrastructure, platform/cloud presence          | I1 ITU IDI; I2 broadband; I3 secure servers; I5 ND-GAIN readiness                                                           |

**Table 2:** Per-vector across-axis weights (each row sums to 1), translating the working paper’s §2.2–§2.4 arguments. Used only for the vulnerability overlay (Section 5); the headline composite does not depend on them.

| Vector                      | Capacity | Governance | Infrastructure |
|-----------------------------|----------|------------|----------------|
| Cyber (§2.2)                | 0.40     | 0.25       | 0.35           |
| CBRN (§2.3)                 | 0.50     | 0.40       | 0.10           |
| Influence operations (§2.4) | 0.20     | 0.35       | 0.45           |

## 2 Conceptual framework

### 2.1 The three axes

Table 1 reproduces the paper’s axis definitions and lists the indicators assigned to each. The fifteen indicators comprise five *capacity-depth* measures, six *governance-orientation* measures, and four *infrastructure-posture* measures.

### 2.2 The three attack vectors

The paper’s §2 argues that each attack vector loads the three axes differently. We encode this as a per-vector set of across-axis weights (Table 2); these weights are the authors’ translation of the paper’s §2.2–§2.4 qualitative arguments, not externally elicited, and their effect on the per-vector ranking is stress-tested in Section 5. Cyber threats emphasise defensive capacity and monitoring infrastructure; CBRN threats emphasise capacity and biosecurity governance; influence operations emphasise open distribution infrastructure and democratic-resilience governance.

### 2.3 Why a composite, and its limits

Composite indicators are routinely criticised for their dependence on weighting. Following [OECD and European Commission, Joint Research Centre \(2008\)](#), our defence is not to claim the weights are uniquely correct, but to (i) report three weighting schemes side by side, (ii) quantify rank stability under Monte Carlo perturbation, (iii) make the indicator-to-axis mapping fully transparent, and (iv) maintain end-to-end source traceability. The handbook is explicit that a composite should

never be presented as a single number without accompanying sensitivity ranges; we adhere to this throughout.

## 3 Data

### 3.1 Indicator set and provenance

Table 3 lists the fifteen indicators with their source, pre-normalisation transform, and within-axis literature weight. Twelve of the fifteen are fully sourced from Tier-1 institutional providers with no analyst-coding step (eleven retrieved programmatically; one—Australia Group membership, G6—inlined against its cited public roster because the upstream page is unreliable for automated fetch). Programmatic sources include Epoch AI (Epoch AI, 2024) for notable-model counts and training compute (C1, C2), OpenAlex (Priem et al., 2022) for AI publications (C3), the Stanford AI Index (Maslej et al., 2025) for AI patents per 100,000 inhabitants (C4), the ITU Global Cybersecurity Index (International Telecommunication Union, 2024a) (G2), V-Dem (Coppedge et al., 2026) for the liberal-democracy index (G3), the IGSC roster (International Gene Synthesis Consortium, 2026) (G5), the ITU ICT Development Index (International Telecommunication Union, 2024b) (I1), the World Bank WDI (World Bank, 2024) for broadband and secure servers (I2, I3), and the ND-GAIN Country Index (University of Notre Dame, 2026) (I5).

Three indicators carry an analyst-coding step and are flagged `requires_verification` in the project’s data registry: G1 (national AI-strategy comprehensiveness, a 0–3 rubric over the OECD.AI Policy Observatory (OECD, 2024)), G4 (bilateral frontier-lab MoU count, an enumeration of public announcements), and the ordinal extension of C5 (AISII presence; the binary 0/3 split between Network and non-Network members is authoritative from the NIST fact sheet (U.S. Department of Commerce, 2024), while the intermediate codes for non-members are authorial). For each, per-country coding protocols and evidence citations are recorded in the notebook, and Section 5 shows that collapsing each indicator to its authoritative core leaves the ranking unchanged.

### 3.2 Missingness and observability

The raw  $14 \times 15$  matrix is 91.9% observed (193 of 210 cells). Figure 1 maps every cell to one of three states: observed, stage-1 axis-mean imputed, or stage-2 column-mean fallback. The largest gap is C4 (AI patents), missing for seven of the fourteen jurisdictions because they fall below the Stanford AI Index’s published top-15 cut-off; C4 therefore exceeds the imputation threshold and its missing cells are column-mean filled. Taiwan’s four infrastructure indicators are unobserved (a missing-not-at-random exclusion from ITU and World Bank ICT collection) and are likewise column-mean filled—a caveat carried through every downstream figure.

## 4 Methodology

The pipeline follows the OECD/JRC ten-step process (OECD and European Commission, Joint Research Centre, 2008): theoretical framework (Section 2), data selection (Section 3), imputation, multivariate analysis, normalisation, weighting, aggregation, robustness, back-to-the-data interpretation, and visualisation.

**Table 3:** The fifteen indicators: axis assignment, source, pre-normalisation transform, and within-axis weight under the literature scheme. Polarity is “higher is better” throughout.

| ID | Indicator (source)                      | Axis           | Transform | Lit. wt. |
|----|-----------------------------------------|----------------|-----------|----------|
| C1 | Notable models, 5-yr (Epoch AI)         | Capacity       | log1p     | 0.30     |
| C2 | Training compute, 5-yr (Epoch AI)       | Capacity       | log1p     | 0.25     |
| C3 | AI publications (OpenAlex)              | Capacity       | log1p     | 0.10     |
| C4 | AI patents per 100k (Stanford AI Index) | Capacity       | log1p     | 0.10     |
| C5 | AISI presence (NIST fact sheet)         | Capacity       | none      | 0.25     |
| G1 | National AI strategy (OECD.AI; coded)   | Governance     | none      | 0.20     |
| G2 | Global Cybersecurity Index (ITU)        | Governance     | none      | 0.30     |
| G3 | Liberal-democracy index (V-Dem)         | Governance     | none      | 0.20     |
| G4 | Bilateral lab-MoU count (coded)         | Governance     | none      | 0.10     |
| G5 | IGSC member firms (IGSC roster)         | Governance     | none      | 0.10     |
| G6 | Australia Group membership (roster)     | Governance     | none      | 0.10     |
| I1 | ICT Development Index (ITU IDI)         | Infrastructure | none      | 0.40     |
| I2 | Fixed broadband per 100 (World Bank)    | Infrastructure | none      | 0.20     |
| I3 | Secure servers per million (World Bank) | Infrastructure | log1p     | 0.20     |
| I5 | ND-GAIN readiness (Notre Dame)          | Infrastructure | none      | 0.20     |

#### 4.1 Cleaning, transforms, and imputation

Count-like indicators (C1–C4, I3) receive a  $\log(1+x)$  transform; bounded scores and ordinals are left untransformed. The European Union row is constructed from member states—summed for count indicators that upstream sources report per country, and member-mean for bounded scores and per-capita rates—a deliberate but internally asymmetric construction that we flag as a limitation (Section 7). Missing cells whose indicator has at most 30% missingness are imputed by the country’s average  $z$ -position across its other same-axis indicators (stage 1); indicators above that threshold, and cells with no observed axis-mate, fall through to column-mean imputation (stage 2). Only six cells receive stage-1 imputation.

#### 4.2 Normalisation and aggregation

Each indicator is normalised by  $z$ -score across the fourteen jurisdictions,

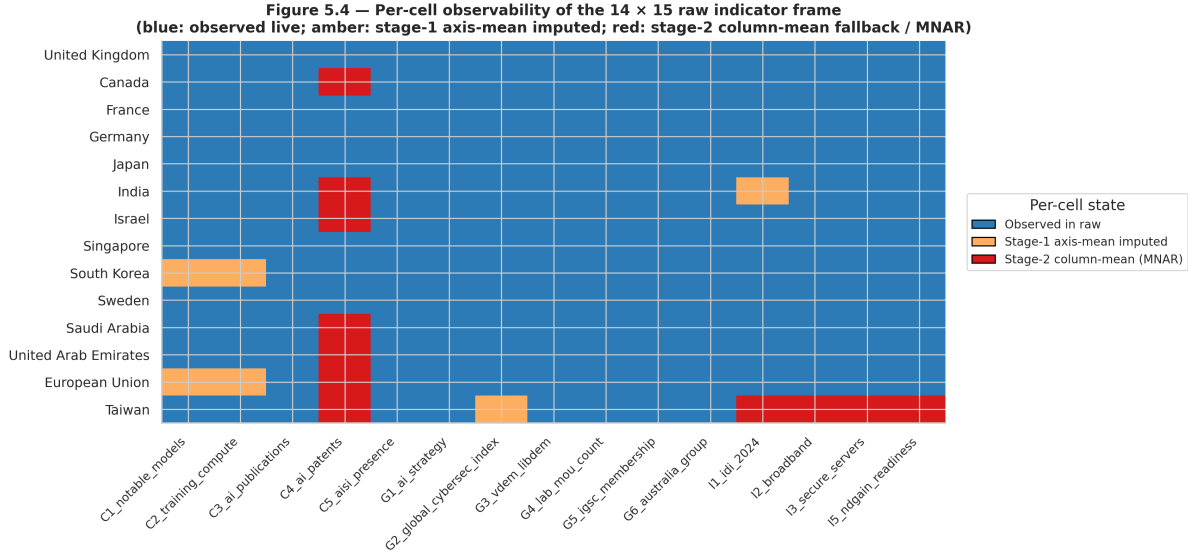
$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j}, \quad (1)$$

with min–max normalisation retained as a sensitivity check. Within each axis  $a$ , the axis score is a linear weighted sum of its normalised indicators,

$$A_{ia} = \sum_{j \in a} w_j z_{ij}, \quad \sum_{j \in a} w_j = 1. \quad (2)$$

Across axes, the composite is a weighted *geometric* mean, which penalises imbalance so that no single axis fully compensates for weakness on another—consistent with the paper’s §4 logic that all three axes matter. Because  $z$ -scored axis scores can be negative, each axis is shifted to positive support before the logarithm:

$$C_i = \exp \left( \sum_a W_a \log(A_{ia} - \min_k A_{ka} + \varepsilon) \right), \quad \sum_a W_a = 1, \quad (3)$$



**Figure 1:** Per-cell observability of the  $14 \times 15$  raw indicator matrix. Blue: observed; amber: stage-1 axis-mean imputed; red: stage-2 column-mean fallback (missing-not-at-random). C4 patents (seven jurisdictions) and Taiwan’s four infrastructure cells dominate the red region. Source: `outputs/missingness_audit.csv`.

with  $\varepsilon = 10^{-3}$ . The shift makes per-country composite *values* comparable only within the fourteen-country frame; the *ranking* is the portable quantity, and the sensitivity of the bottom tier to  $\varepsilon$  is quantified in Section 5.

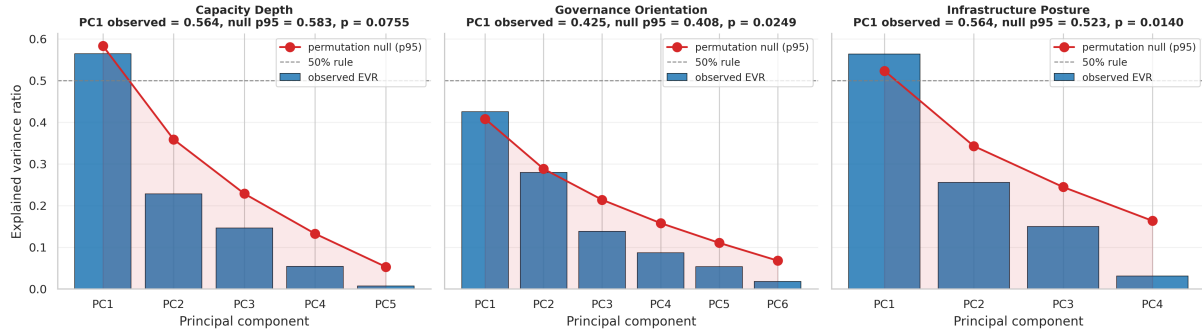
### 4.3 Weighting schemes

Three schemes are reported side by side: **equal** (uniform within and across axes); **PCA-derived** (within-axis weights proportional to squared first-principal-component loadings, computed at run time); and **literature-elicited** (Table 3; across-axis weights 0.40/0.30/0.30), synthesised from the GMF *Pivotal Powers* (de Hoop Scheffer et al., 2024), Chatham House sovereign-AI (Sandoval et al., 2026), and Tortoise Global AI Index (Tortoise Media, 2023) frameworks. The PCA scheme is presented as a sensitivity benchmark only: at  $n = 14$  its loadings are unstable, and on the governance axis it effectively zeroes the cybersecurity indicator (G2) and reallocates mass to G1.

### 4.4 Multivariate diagnostics

We assess whether each axis coheres as a one-dimensional construct using within-axis correlations, Cronbach’s  $\alpha$  (Cronbach, 1951), and Horn’s parallel analysis (Horn, 1965) (Figure 2). Parallel analysis compares each axis’s observed first-component explained-variance ratio against a 95th-percentile permutation null over 10,000 shuffles. The infrastructure and governance axes clear the null ( $p = 0.014$  and  $p = 0.025$ ); the capacity axis is borderline ( $p = 0.076$ ) and, with only seven complete cases, is effectively untestable. Cronbach’s  $\alpha$  is 0.741 (capacity), 0.663 (governance, below the conventional 0.7 threshold), and 0.722 (infrastructure).

Figure 8.1 — Horn’s parallel analysis per axis (N = 10,000 permutations)



**Figure 2:** Horn’s parallel analysis per axis (10,000 permutations). Bars: observed explained-variance ratio per component; red line: 95th-percentile permutation null; dashed grey: the 50% heuristic. The first component clears the null for governance and infrastructure but not for capacity (fit on  $n = 7$  complete cases). Source: `outputs/parallel_analysis.csv`.

#### 4.5 Sensitivity, robustness, and inference

Rank stability is probed with: a 10,000-draw Dirichlet( $\alpha=1$ ) Monte Carlo over within- and across-axis weights; leave-one-indicator-out (LOO); a  $z$ -score-versus-min-max normalisation swap; a bibliographic-classifier swap (broad-AI versus narrow-ML OpenAlex concept for C3); an  $\varepsilon$ -grid; a C5-axis-reassignment check; and an analyst-coded judgment-collapse check. All rank correlations are reported with Bonett–Wright (Bonett and Wright, 2000) Fisher- $z$  95% confidence intervals, whose standard error

$$SE = \sqrt{\frac{1 + \rho^2/2}{n - 3}} \quad (4)$$

is the variance-stabilised form appropriate for Spearman’s  $\rho$  (wider than the naive Pearson  $1/\sqrt{n-3}$ ). A second, *joint* Monte Carlo perturbs the imputed cells alongside the weights—stage-1 axis-mean cells by the country’s observed within-axis  $z$ -dispersion, and stage-2/MNAR cells by  $N(0, \sigma_2)$  with  $\sigma_2 \in \{0.5, 1.0\}$ —while holding every *observed* value fixed, yielding per-country credible rank intervals and tier-membership probabilities (Section 5, Figure 5). Concurrent validity is assessed against an independent external index (Section 6.1).

#### 4.6 Typology and interpretability

Country archetypes are obtained by  $k$ -means with a silhouette-selected  $k$  (Rousseeuw, 1987), cross-checked against hierarchical clustering (adjusted Rand index, Hubert and Arabie, 1985) and a multidimensional-scaling embedding (Kruskal, 1964). Per-indicator contributions to the headline composite are decomposed exactly by enumerating all  $2^{15}$  Shapley (Shapley, 1953) coalitions, complemented by permutation feature importance and, over the same coalition set, exact second-order Shapley *interaction* indices (Grabisch and Roubens, 1999) that quantify pairwise indicator redundancy (Appendix C, Figure 19). Internal consistency, clustering, and MDS use `scikit-learn` (Pedregosa et al., 2011); the stack also includes NumPy (Harris et al., 2020), pandas (McKinney, 2010), SciPy (Virtanen et al., 2020), Matplotlib (Hunter, 2007), and seaborn (Waskom, 2021).



**Table 4:** Headline composite scores under the three weighting schemes, ordered by the literature scheme. The top-five set is invariant across schemes. Source: `outputs/index_baseline.csv`.

| Rank | Country              | Equal | PCA   | Literature |
|------|----------------------|-------|-------|------------|
| 1    | United Kingdom       | 2.227 | 2.571 | 2.314      |
| 2    | South Korea          | 1.951 | 1.998 | 2.185      |
| 3    | France               | 1.792 | 2.041 | 1.980      |
| 4    | European Union       | 1.785 | 2.126 | 1.824      |
| 5    | Japan                | 1.725 | 1.897 | 1.823      |
| 6    | Singapore            | 1.418 | 1.549 | 1.809      |
| 7    | Canada               | 1.446 | 1.837 | 1.472      |
| 8    | Germany              | 1.285 | 1.335 | 1.434      |
| 9    | Sweden               | 0.958 | 0.978 | 1.157      |
| 10   | United Arab Emirates | 0.673 | 0.307 | 1.020      |
| 11   | Saudi Arabia         | 0.628 | 0.116 | 0.922      |
| 12   | Israel               | 0.100 | 0.661 | 0.144      |
| 13   | India                | 0.073 | 0.067 | 0.106      |
| 14   | Taiwan               | 0.065 | 0.060 | 0.057      |

## 5 Results

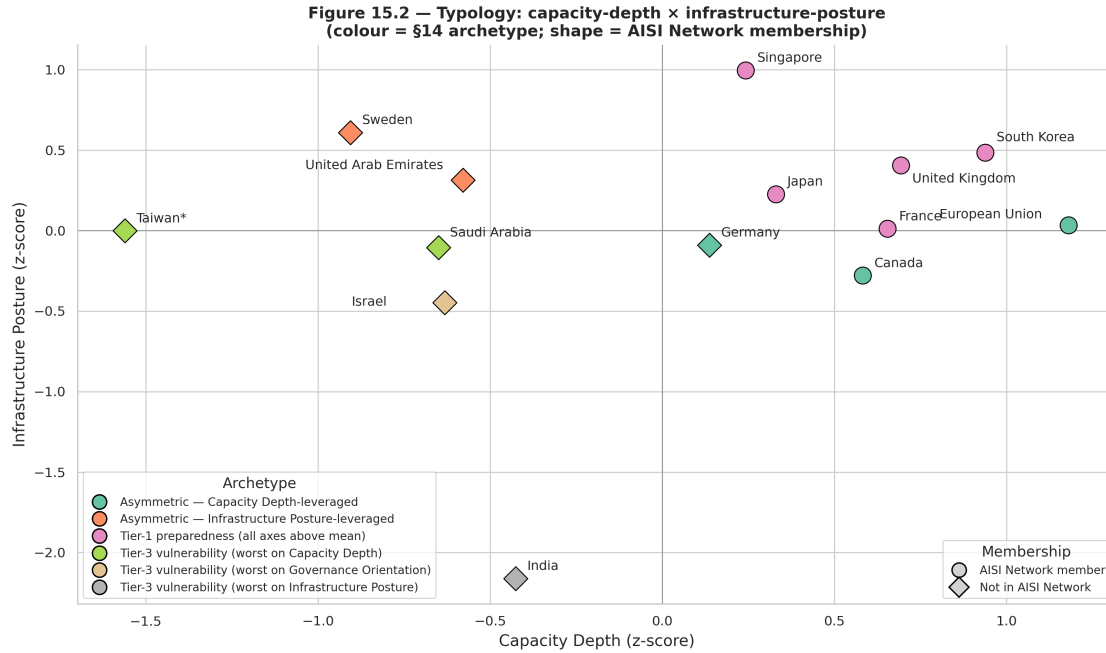
### 5.1 Headline ranking

Table 4 reports the composite under all three schemes. The literature-weighted ranking is led by the United Kingdom (2.314) and South Korea (2.185), and trailed by India (0.106) and Taiwan (0.057). The top-five set {UK, Korea, France, EU, Japan} is identical across all three schemes; the bottom-three set {Israel, India, Taiwan} is identical under the equal and literature schemes (under PCA, Israel rises to rank 10 and Saudi Arabia enters the bottom three). Figure 3 positions the countries on the capacity  $\times$  infrastructure plane, colour-coded by archetype and shape-coded by AISI Network membership.

### 5.2 Robustness

Under the Bonett–Wright standard error, the three weighting/normalisation perturbations clear the 0.70 robustness floor with wide margins (Table 5, Figure 4). The bibliographic-classifier check has a point estimate of  $\rho = 0.960$  but a 95% CI lower bound of 0.845, which clears the 0.70 floor yet sits marginally below the stricter 0.85 bar imposed on that check (a core measurement choice rather than a downstream weight perturbation). The Monte Carlo rank distributions (Appendix Figure 10) show the United Kingdom tightest at rank 1 and the {France, EU, Japan} block sharing a median rank of 4.

**Rank uncertainty.** The joint Monte Carlo—perturbing weights *and* imputed cells while holding observed values fixed—attaches an uncertainty band to every position. Under the most conservative setting ( $\sigma_2 = 1.0$ ), the United Kingdom holds rank 1 in 99% of draws and South Korea sits in the top five in 92%; the bottom tier is comparably stable (India and Taiwan in the bottom three in 82% and 78% of draws). The added imputation noise widens the credible intervals only marginally over



\* Taiwan's infrastructure-posture coordinate is column-mean-imputed (four infrastructure indicators are unobserved for Taiwan; see §14 observability caveat).

**Figure 3:** Country positions on Capacity Depth × Infrastructure Posture ( $z$ -scores). Colour encodes the per-country archetype derived from each country's own axis profile (Tier-1; two Asymmetric variants; three Tier-3 variants by worst axis); marker shape encodes AISI Network membership. Taiwan's infrastructure coordinate is column-mean-imputed. Source: `outputs/typology.csv`.

the weight-only Monte Carlo (Figure 5), confirming that the headline ordering is not an artefact of the fixed imputed values—the cells the pipeline had to fill do not drive the ranking. The middle of the table is where the bands overlap most, consistent with the single-slot instability behind H6's partial verdict.

### 5.3 Pre-registered hypotheses

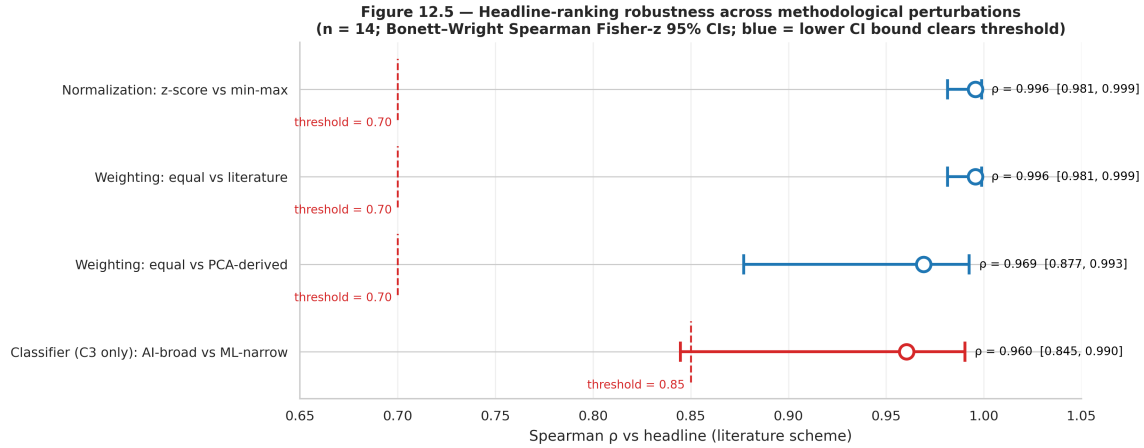
Table 6 records the six hypotheses, their numeric pass criteria, and their mechanically resolved verdicts. Two are fully supported (H3, H5) and four are partially supported with named caveats (H1, H2, H4, H6). The partial verdicts surface specific methodological caveats—a near-collinear pair across two axes (C5×G1, Pearson  $r = 0.973$ ), governance multidimensionality, the borderline classifier check, and single-slot tier cycling—rather than a wholesale framework failure. The C5×G1 collinearity is the largest of only five cross-axis indicator pairs at  $|r| \geq 0.7$  (of 74 pairs).

### 5.4 Vulnerability overlay

Applying the per-vector weights (Table 2) to the literature axis scores yields per-country preparedness scores on each attack vector (Figure 6; full ranks in Appendix Table 7). The United Kingdom and South Korea lead on all three vectors; the lower tier (Israel, Taiwan, India) is least prepared, with vector-specific reordering—India is weakest on cyber and influence operations, Taiwan weakest on CBRN. Per-vector scores are signed  $z$ -scale quantities (positive = above sample-mean preparedness) and are interpretable only within a vector, not across vectors.

**Table 5:** Robustness of the headline ranking. Spearman  $\rho$  against the literature ranking with Bonett–Wright Fisher- $z$  95% CIs ( $n = 14$ ). Source: `outputs/robustness_summary_with_ci.csv`.

| Perturbation                      | $\rho$ | Threshold | 95% CI         | Verdict    |
|-----------------------------------|--------|-----------|----------------|------------|
| Weighting: equal vs PCA           | 0.969  | 0.70      | [0.877, 0.993] | clears     |
| Weighting: equal vs literature    | 0.996  | 0.70      | [0.982, 0.999] | clears     |
| Normalisation: $z$ vs min-max     | 0.996  | 0.70      | [0.982, 0.999] | clears     |
| Classifier: AI-broad vs ML-narrow | 0.960  | 0.85      | [0.845, 0.990] | borderline |

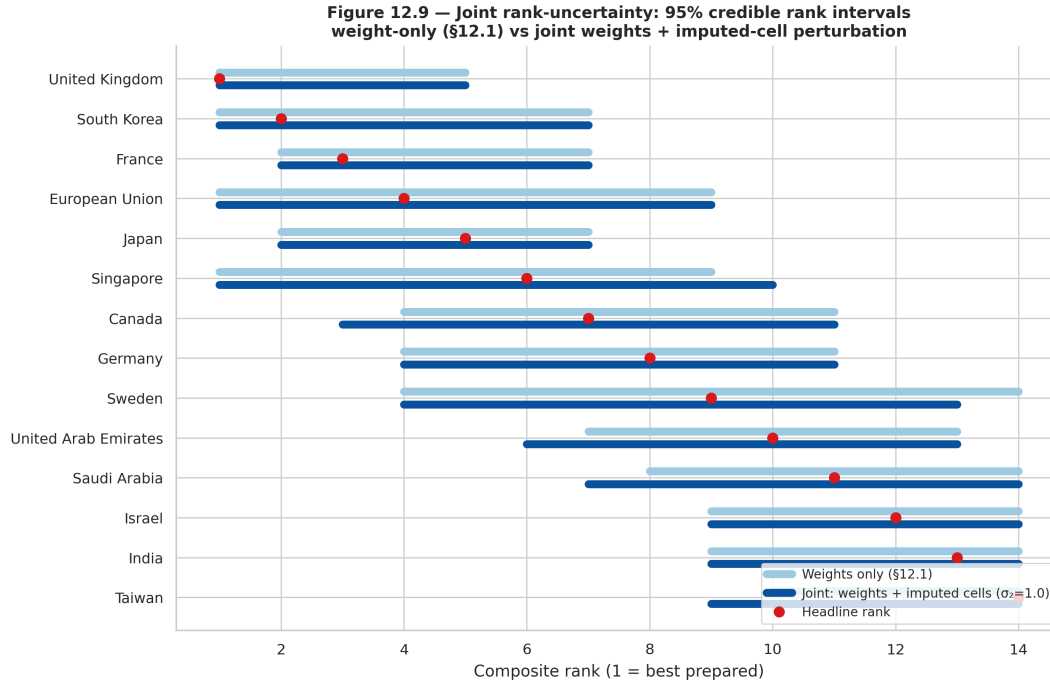
**Figure 4:** Robustness forest plot: Spearman  $\rho$  point estimates with Bonett–Wright Fisher- $z$  95% CIs against each perturbation’s pass threshold (dashed). The three weighting/normalisation checks clear 0.70 with wide margins; the classifier-sensitivity check is borderline against its stricter 0.85 bar. Source: `outputs/robustness_summary_with_ci.csv`.

## 5.5 Counterfactual policy leverage

To operationalise the paper’s §4 claim that tractability depends on axis position, we recompute the composite under five direct policy actions per country (e.g. joining the AISI Network, publishing a safety-provisioned AI strategy, joining the Australia Group). Figure 7 reports the composite change; the highest-leverage action differs by country and by starting position. For example, with Singapore selected, joining the Australia Group raises its composite from 1.809 to 1.911 (rank 6 to 4). For countries at an axis minimum, the geometric-shift artefact can understate the composite change, so an axis-level reading is reported alongside.

## 5.6 Typology

The silhouette-selected  $k = 3$   $k$ -means partition separates a Tier-1/asymmetric-developed group, a mid group, and India as a singleton; the partition is identical to Ward and average-linkage hierarchical clustering (ARI = 1.0; complete linkage 0.733). The MDS embedding (Figure 8) preserves the pairwise-distance rank at  $\rho = 0.907$  and visually separates the archetypes.



**Figure 5:** Joint rank-uncertainty: 95% credible rank intervals from 10,000 draws, weight-only (Section 4, §12.1) versus joint weights-plus-imputed-cell perturbation ( $\sigma_2 = 1.0$ ); red markers are the headline ranks. Observed Tier-1 values are held fixed; only imputed cells are resampled. Source: `outputs/rank_uncertainty.csv`.

## 6 Discussion: how the index supports the working paper

The index supplies the empirical layer the working paper asserts but does not provide.

**It grounds the §1 premise.** The paper claims the fourteen middle powers “differ along three axes.” H3 confirms this empirically: the countries do not cluster homogeneously (silhouette 0.426; a clustering that is identical across two linkage methods), and the typology identifies coherent archetypes—Tier-1 (UK, Korea, France, Japan, Singapore), capacity-leveraged (EU, Canada, Germany), infrastructure-leveraged (Sweden, UAE), and a Tier-3 vulnerable group (Israel, Saudi Arabia, India, Taiwan).

**It identifies the most-exposed tier.** The paper argues that lower-capacity middle powers face a “structurally under-addressed vulnerability.” The index makes that tier concrete and reproducible: India and Taiwan are the two lowest-ranked composites under all three weighting schemes, and the three least-prepared states on every per-vector overlay are India, Israel, and Taiwan (in vector-specific order); the third bottom-tier composite slot cycles between Israel and Saudi Arabia (the single-slot instability behind H6’s partial verdict). This illustrates—rather than merely asserts—which states the paper’s argument most urgently concerns.

**It quantifies §4 tractability.** The paper repeats that “tractability depends on a country’s position within the three axes” without specifying any action’s leverage. The counterfactual

**Table 6:** The six pre-registered hypotheses (criteria fixed before estimation) and their mechanically resolved verdicts.

| ID | Hypothesis                                          | Verdict   | Evidence                                                                                                                                |
|----|-----------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| H1 | The three axes are empirically separable.           | Partial   | 5 of 74 cross-axis pairs at $ r  \geq 0.7$ ; C5×G1 ( $r = 0.973$ ) is the largest contributor.                                          |
| H2 | Each axis carries a one-dimensional signal.         | Partial   | Infrastructure $p = 0.014$ , governance $p = 0.025$ (both significant); capacity untestable at $n = 7$ ; governance PC1 explains 42.5%. |
| H3 | The 14 powers do not cluster homogeneously.         | Supported | Silhouette 0.426 at $k = 3$ ; ARI = 1.0 vs Ward and average linkage; MDS distance-rank $\rho = 0.907$ .                                 |
| H4 | The ranking is robust to perturbation.              | Partial   | Weighting/normalisation clear with wide margins; classifier CI lower bound 0.845 is below its 0.85 bar.                                 |
| H5 | The top-5 is not driven by any one coded indicator. | Supported | Top-5 set unchanged dropping any of G1/G4/C5; max single-country shift 2.                                                               |
| H6 | Top-5 and bottom-3 tier membership are stable.      | Partial   | 5/5 top-5 members in tier in $\geq 70\%$ of draws; Israel only 62.2% in the bottom-3, cycling with Saudi Arabia.                        |

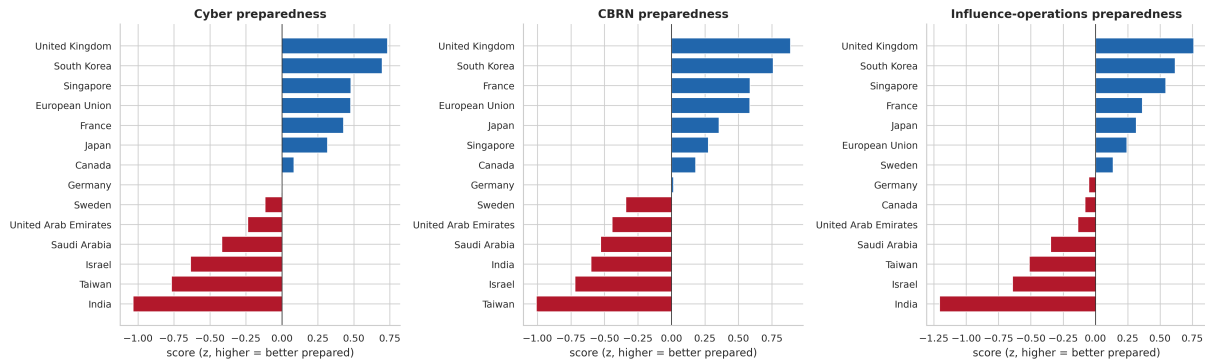
scenarios (Figure 7) convert this into a per-country, per-action quantity: the highest-leverage step differs by starting position, exactly as the paper’s qualitative argument implies.

**Defensibility.** Three design choices keep the study reviewable. First, *bounded scope*: the index operationalises §1–§2 and explicitly declines §3–§4. Second, *census, not sample*: the fourteen jurisdictions are the population the paper names, so the descriptive claims are exact and the small- $n$  multivariate methods are caveated diagnostics rather than load-bearing inference. Third, *transparency*: every weakness is disclosed and bounded by a dedicated sensitivity analysis, and the partial-support verdicts are reported as such.

## 6.1 External concurrent validity

A composite is only as credible as its agreement with *independent* measures of the same construct. None of M-PAPI’s indicators is itself a published readiness score, so the index can be benchmarked against one without circularity. The natural comparator is the Oxford Insights Government AI Readiness Index 2024 (Oxford Insights, 2024), an independently constructed index over 188 jurisdictions and 40 indicators that is *not* an input to M-PAPI. Because it scores sovereign states, it covers thirteen of the fourteen middle powers—the EU bloc is the sole exclusion, mirroring the Section 5 EU-vs-member-state treatment—and Taiwan *is* covered. Across the thirteen, the M-PAPI literature composite and the GAIRI overall score are strongly rank-concordant: Spearman  $\rho = 0.857$  (95% CI [0.506, 0.965],  $n = 13$ ; Figure 9). The agreement evidences that M-PAPI tracks a real, externally corroborated readiness gradient rather than an artefact of its own weighting. The divergences are equally informative and expected by construction: Singapore ranks first on GAIRI but fifth on M-PAPI, and the United Kingdom and Japan rank higher on M-PAPI than on GAIRI—M-PAPI deliberately rewards the CBRN/biosecurity-governance signals (Australia Group membership, IGSC participation, AISI-network presence) that a general-purpose government-readiness index omits, which is precisely the proliferation-specific lens the working paper calls for.

Figure 15.3 — Vulnerability overlay per attack vector (paper §2)



**Figure 6:** Per-vector preparedness (paper §2): cyber, CBRN, and influence operations. Each panel is sorted independently; bars are signed  $z$ -scale scores (higher = better prepared). Source: `outputs/vulnerability_overlay.csv`.

## 7 Limitations and threats to validity

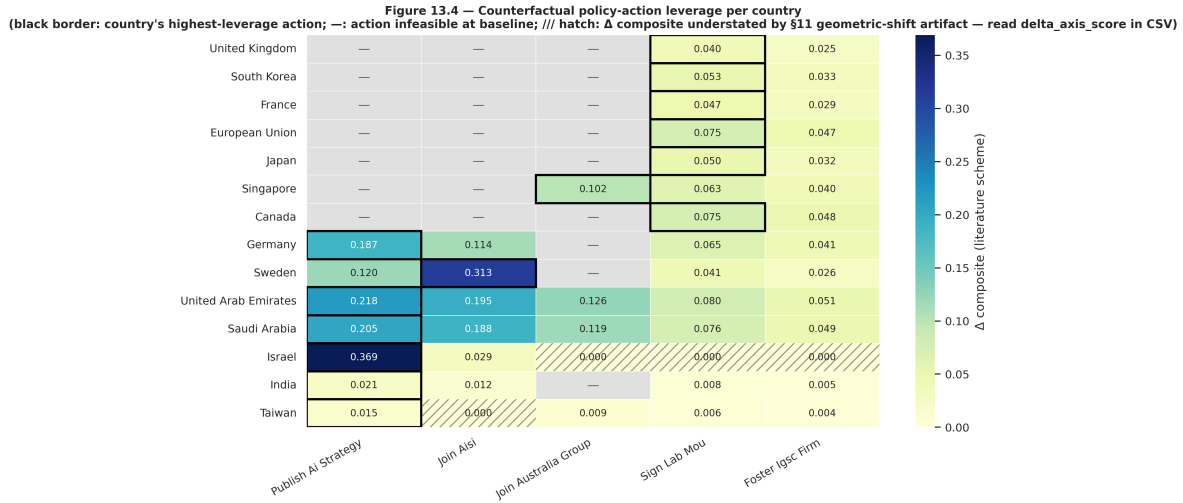
Following [Shadish et al. \(2002\)](#), we organise residual risks by validity type.

**Construct validity.** Several indicators are disclosed proxies whose link to the axis label is indirect: ND-GAIN readiness (I5) is a climate-adaptation measure used as an infrastructure-resilience proxy; V-Dem (G3) measures political-system democracy, not AI governance *per se*; the IGSC count (G5) measures industrial biosecurity participation; and secure-server density (I3) increasingly reflects certificate-registration jurisdiction rather than substantive infrastructure. These are the most pressable choices; the defence is that the axes are the paper’s, the proxies are transparent, and the ranking is robust to dropping any single indicator.

**Cross-axis collinearity.** C5 (AISII presence) and G1 (AI-strategy comprehensiveness) are near-collinear ( $r = 0.973$ ) yet sit in different axes, so a shared institutional signal enters the composite twice. Reassigning C5 to governance leaves the top-five set unchanged (Spearman  $\rho = 0.996$ , maximum single-country shift 1), so the double-count does not re-order the table. The exact second-order Shapley interaction index puts a magnitude on the overlap *inside* the composite:  $I(C5, G1) = +0.008$ —small and mildly *complementary*, not substitutive. The high correlation is thus a shared-information property (the two indicators co-vary across countries) rather than a large additive redundancy in the aggregated score; because C5 and G1 sit in different axes, the cross-axis geometric mean makes their joint marginal contribution slightly positive. The genuinely substitutive interactions are all within-axis pairs (e.g.  $C1 \times C2 = -0.019$ ), exactly as linear within-axis aggregation of correlated indicators predicts (Appendix C). The double-count remains an acknowledged construct flaw—two indicators of one latent institutional-capacity signal—but its measured aggregate effect is bounded.

**Weighting and aggregation.** The vector-overlay weights are authorial translations of the paper’s qualitative arguments, so the per-vector rankings are the most judgment-dependent output. The across-axis geometric mean introduces a shift constant  $\varepsilon$ : an  $\varepsilon$ -grid spanning  $10^{-4}$  to 1 shows the ranking is  $\varepsilon$ -invariant (Spearman  $\rho = 1.000$  for  $\varepsilon \leq 0.1$ , 0.991 at  $\varepsilon = 1$ ), but the bottom-tier composite *values* are an  $\varepsilon$ -artefact (Taiwan’s value spans 0.023 to 1.54 across the grid). The bottom tier should therefore be cited as ranks, not magnitudes.

**Inference and external validity.** With  $n = 14$ , point-estimate rank correlations carry



**Figure 7:** Counterfactual policy-action leverage per country (composite change under the literature scheme). Black border: the country’s highest-leverage action; em-dash: action already satisfied at baseline; hatching: a geometric-shift artefact for which the axis-level change should be read instead. Source: `outputs/counterfactual_scenarios.csv`.

non-negligible sampling variance, which the Bonett–Wright CIs make explicit. The sample is the fourteen named middle powers only; results do not generalise to non-middle-powers or small states, and composite values are not portable outside the fourteen-country reference frame. The ranking does, however, agree strongly with the independent Oxford Insights GAIRI 2024 index over the thirteen jurisdictions both cover (Spearman  $\rho = 0.857$ ; Section 6.1), which bounds the concern that the ordering is an artefact of the authors’ own weighting.

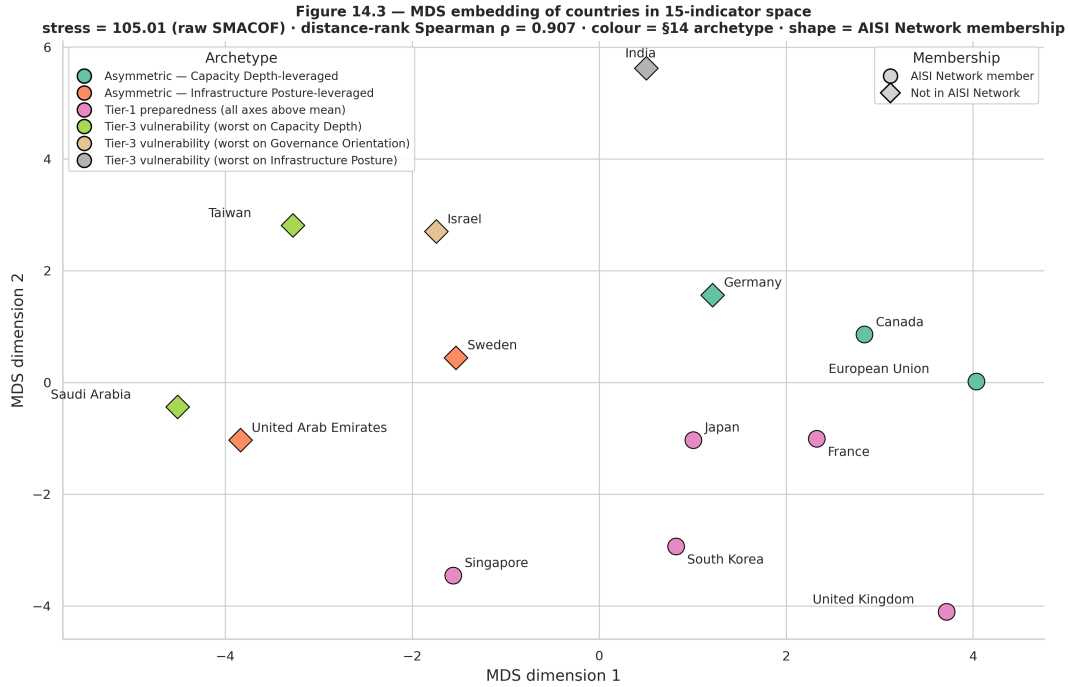
## 8 Conclusion

M-PAPI converts the working paper’s qualitative three-axes framework into a transparent, reproducible composite indicator over the paper’s own fourteen named middle powers. It empirically positions each country, confirms that the countries are differentiable, identifies the most-exposed tier, and quantifies the per-country leverage of preparedness actions—while honestly declining the trigger-event and checklist claims it cannot yet test. The headline ranking is robust to weighting and normalisation and borderline only to a single indicator-construction choice; the hypothesis verdicts (two fully, four partially supported) are reported with their specific caveats; and every output regenerates byte-for-byte. The study is offered as auditable empirical scaffolding for the working paper, and as a template that can be re-run as the underlying data and the threat landscape evolve.

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**Figure 8:** Multidimensional-scaling embedding of the 14 jurisdictions in the full fifteen-indicator space (distance-rank  $\rho = 0.907$ ). Colour: archetype; marker shape: AISI Network membership. Source: `outputs/mds_coordinates.csv`.

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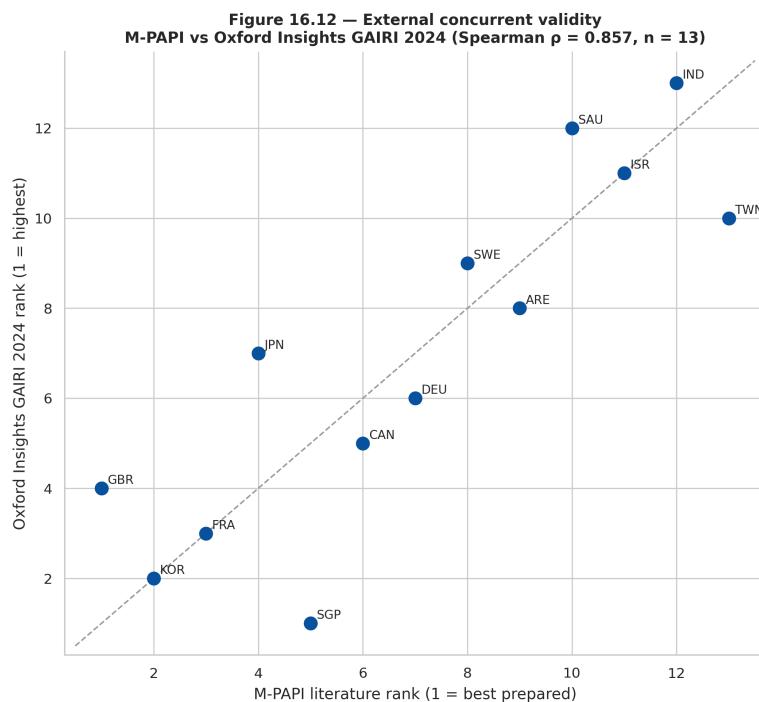
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**Figure 9:** External concurrent validity: M-PAPI literature rank versus Oxford Insights GAIRI 2024 rank across the thirteen covered jurisdictions (the EU bloc is unscored by GAIRI). Points near the diagonal agree; Spearman  $\rho = 0.857$ . Source: `outputs/external_validity.csv`.

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**Table 7:** Per-vector preparedness scores and ranks. Higher score = better prepared; signed  $z$ -scale, interpretable within a vector only. Source: `outputs/vulnerability_overlay.csv`.

| Country              | Cyber | Rk | CBRN  | Rk | Infl. | Rk |
|----------------------|-------|----|-------|----|-------|----|
| United Kingdom       | 0.73  | 1  | 0.89  | 1  | 0.76  | 1  |
| South Korea          | 0.70  | 2  | 0.76  | 2  | 0.62  | 2  |
| Singapore            | 0.48  | 3  | 0.27  | 6  | 0.54  | 3  |
| European Union       | 0.48  | 4  | 0.58  | 4  | 0.24  | 6  |
| France               | 0.43  | 5  | 0.59  | 3  | 0.36  | 4  |
| Japan                | 0.32  | 6  | 0.36  | 5  | 0.32  | 5  |
| Canada               | 0.08  | 7  | 0.18  | 7  | -0.08 | 9  |
| Germany              | -0.00 | 8  | 0.02  | 8  | -0.05 | 8  |
| Sweden               | -0.12 | 9  | -0.34 | 9  | 0.14  | 7  |
| United Arab Emirates | -0.24 | 10 | -0.44 | 10 | -0.14 | 10 |
| Saudi Arabia         | -0.42 | 11 | -0.53 | 11 | -0.35 | 11 |
| Israel               | -0.63 | 12 | -0.72 | 13 | -0.64 | 13 |
| Taiwan               | -0.77 | 13 | -1.01 | 14 | -0.51 | 12 |
| India                | -1.03 | 14 | -0.60 | 12 | -1.21 | 14 |

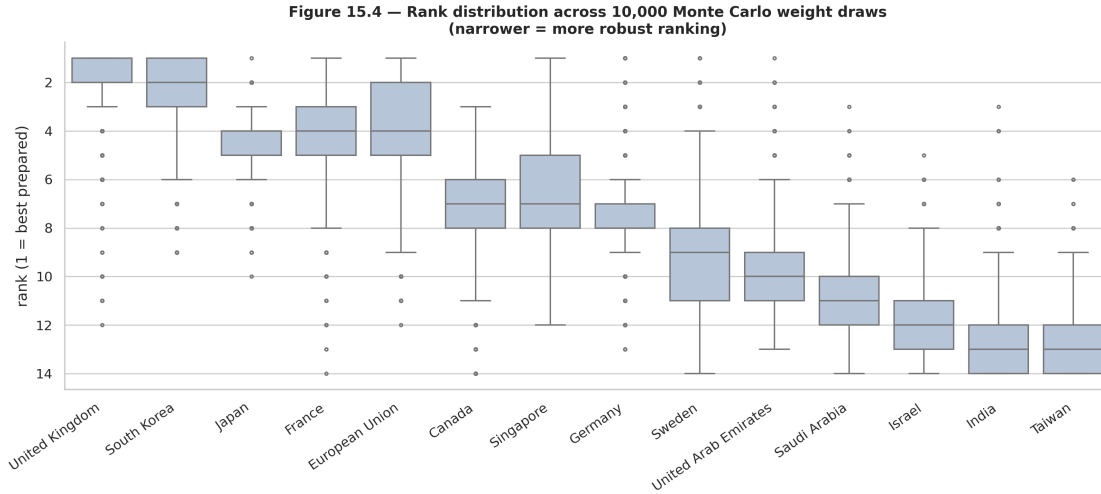
## A Data dictionary and provenance

Every indicator value cites a named source with a retrieval date and URL, recorded in per-file `*.meta.json` provenance sidecars. Programmatic API sources were retrieved on 2026-05-06; the PDF/HTML-derived sources (Stanford patents, ITU IDI, IGSC roster) on 2026-05-18. The bundled `data/raw/` snapshot makes the analysis reproducible offline. Source licences: Epoch AI (CC-BY 4.0), World Bank (CC-BY 4.0), OpenAlex (CC-0), V-Dem (CC-BY), ND-GAIN (free with attribution), ITU (citabile academic use), Stanford HAI (free with attribution), NIST AISI Network (U.S. public domain), IGSC and Australia Group (public rosters).

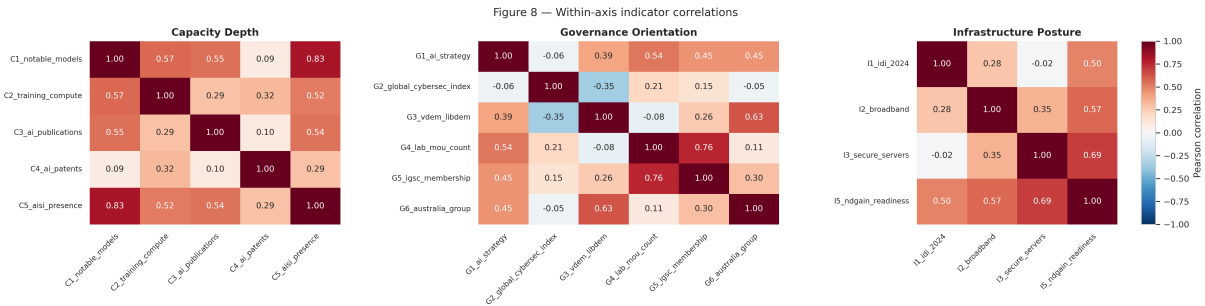
## B Full robustness and dimensionality

Table 7 gives the full per-vector vulnerability ranking. Figure 10 gives the Monte Carlo rank distributions; Figure 11 the within-axis indicator correlations; Figure 12 the full indicator correlation matrix; Figure 13 the hierarchical clustering under three linkage methods; Figures 14, 15, and 16 the axis scoreboard, pairwise axis scatter matrix, and sparse-PCA governance sub-axes respectively.

The  $\varepsilon$ -sensitivity, C5-reassignment, and analyst-coded judgment-collapse checks all leave the top-five set intact: the analyst-collapse variants (G1, C5, G4 individually and all three at once) each return Spearman  $\rho = 1.000$  with zero rank shift, and the C5-reassignment returns  $\rho = 0.996$ . Cross-axis correlations at  $|r| \geq 0.7$  number five of seventy-four pairs (C5×G1 0.973; C1×G1 0.779; C4×I3 -0.771; G3×I3 0.722; C3×G6 0.702). Tier-membership stability over 10,000 Monte Carlo draws: top-5 members in tier  $\geq 80\%$  of draws (UK 99.0%, Korea 93.2%, France 83.8%, Japan 82.5%, EU 80.5%); bottom-3 members India 82.0%, Taiwan 81.8%, Israel 62.2%; exact-set-match 46%/43% and within-one-swap 94%/83% for the top-5/bottom-3.



**Figure 10:** Per-country rank distribution across 10,000 Monte Carlo weight draws (narrower = more robust). Source: `outputs/sensitivity_ranks.csv`.



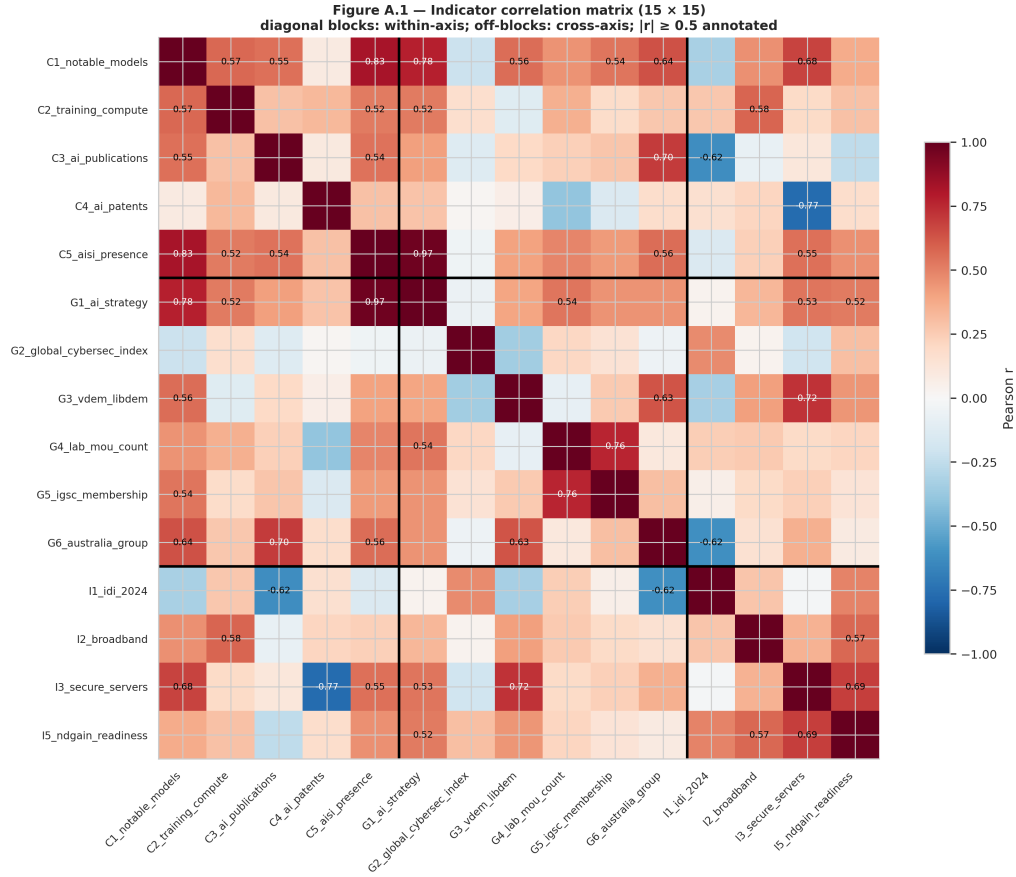
**Figure 11:** Within-axis indicator correlations (Pearson) for the three axes, with a shared colour scale. The governance panel shows the weak/negative association of the ITU GCI (G2) with its axis-mates that underlies the governance axis’s lower internal consistency. Source: `outputs/cross_axis_correlations.csv`.

## C Interpretability

Figure 17 decomposes each country’s composite into exact Shapley contributions per indicator (all  $2^{15}$  coalitions; additivity verified to  $10^{-15}$ ). Figure 18 reports permutation feature importance for the literature ranking. Training compute (C2) and notable-model count (C1) are the most rank-disruptive indicators; ND-GAIN readiness (I5) and patents (C4) the least. Figure 19 reports the exact second-order Shapley interaction indices (Grabisch and Roubens, 1999) for every indicator pair, averaged over the fourteen countries: the within-axis pairs are the substitutive (negative) ones, while cross-axis pairs—including the much-discussed  $C5 \times G1$  (+0.008)—are weakly complementary, locating indicator redundancy firmly within rather than across axes.

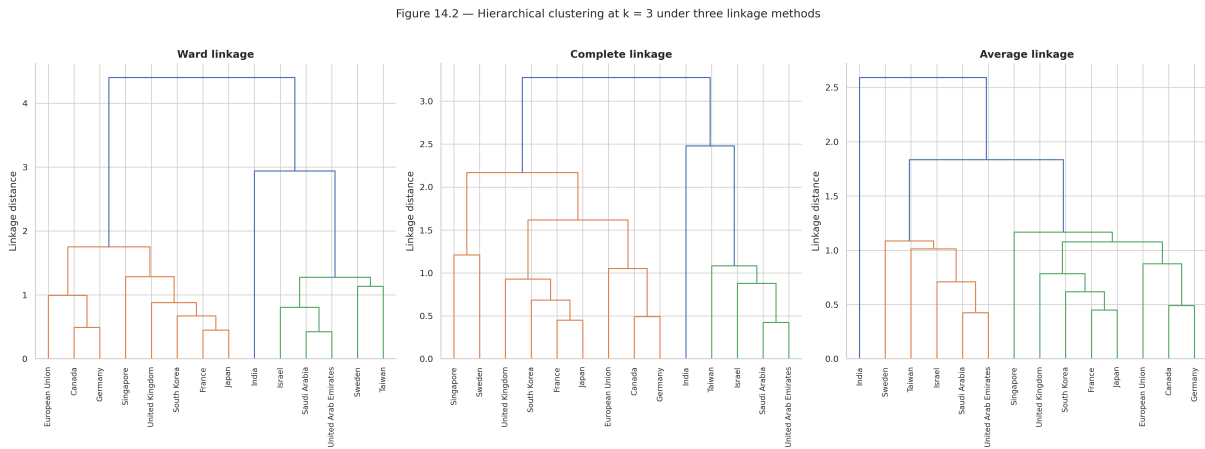
## D Reproducibility

All stochastic steps are seeded (`SEED = 20260506`): the Monte Carlo weights,  $k$ -means, MDS, and sparse PCA. With a pinned environment and the offline data snapshot (`MPAPI_OFFLINE=1`), the twenty-eight output files (twenty-five CSV, three JSON) and all nineteen figures reproduce

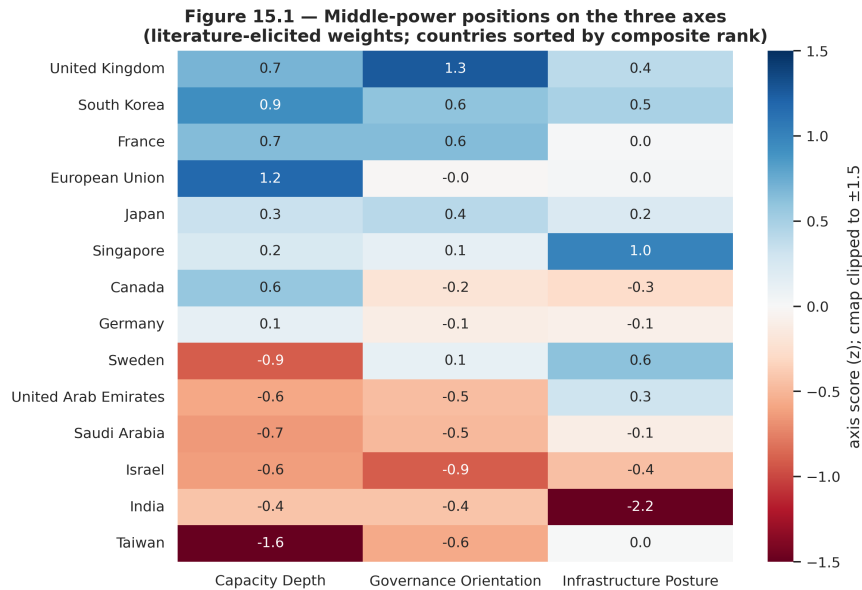


**Figure 12:** Full  $15 \times 15$  indicator correlation matrix with axis-block separators;  $|r| \geq 0.5$  annotated. Diagonal blocks are within-axis; off-blocks are cross-axis. Source: `outputs/cross_axis_correlations.csv`.

byte-for-byte across successive executions, verified by hashing two consecutive offline runs. The notebook additionally reconstructs one country’s composite end-to-end (asserted within  $10^{-3}$ ) and verifies Shapley additivity (within  $10^{-6}$ ). The analysis is built on the open scientific Python stack (Harris et al., 2020; McKinney, 2010; Virtanen et al., 2020; Pedregosa et al., 2011; Hunter, 2007; Waskom, 2021).

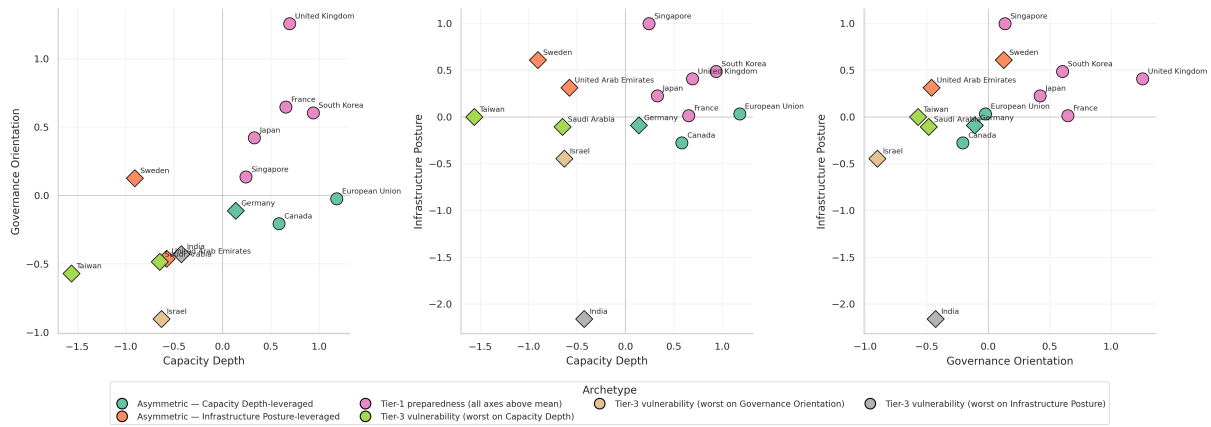
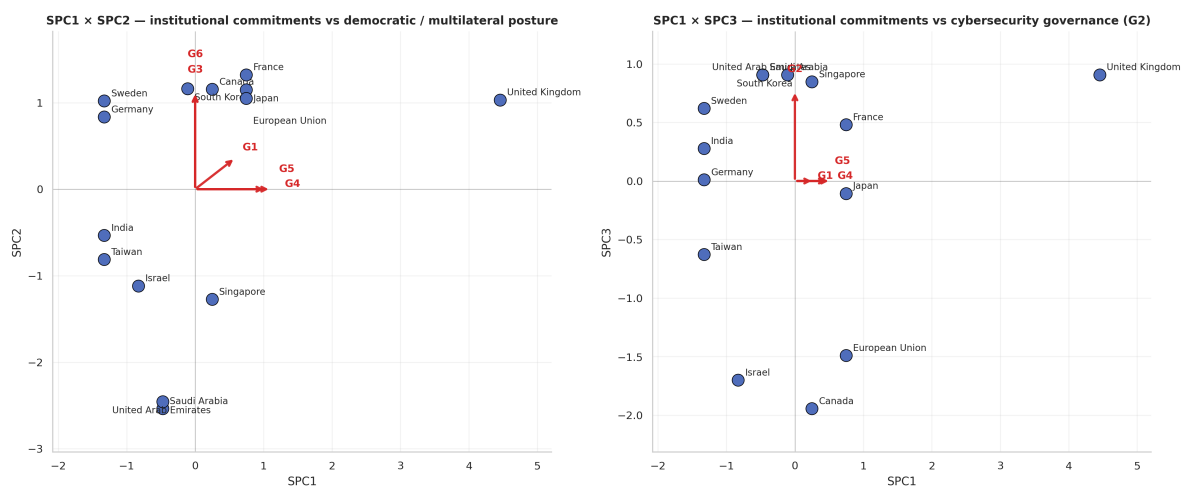


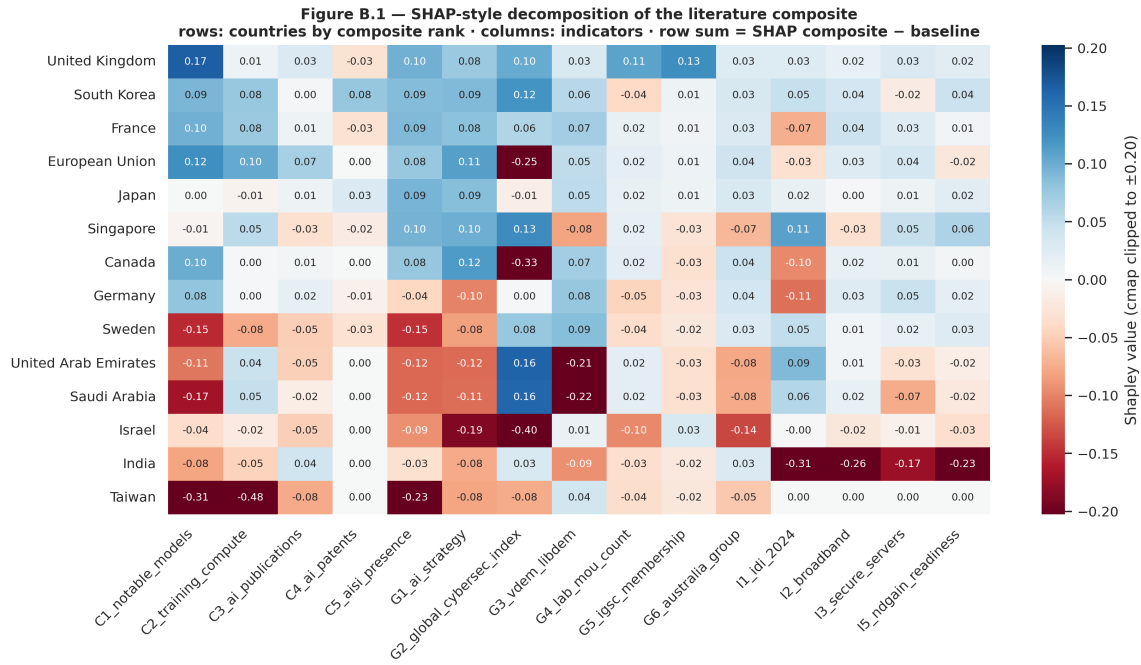
**Figure 13:** Hierarchical clustering at  $k = 3$  under Ward, complete, and average linkage. Ward and average partitions are identical to the  $k$ -means partition (ARI = 1.0). Source: `outputs/hierarchical_ari.csv`.



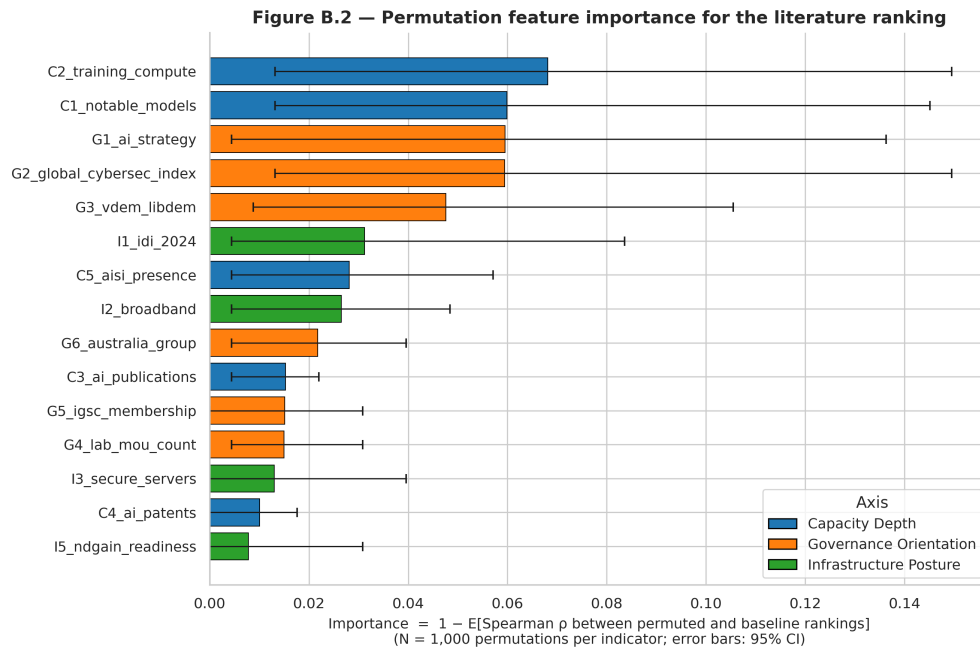
Taiwan's Infrastructure Posture cell is column-mean-imputed (its four infrastructure indicators are unobserved; see §14 observability caveat).

**Figure 14:** Per-country axis scores under the literature scheme ( $z$ -scale, clipped to  $\pm 1.5$ ), ordered by composite rank. Taiwan's infrastructure cell is column-mean-imputed. Source: `outputs/index_baseline.csv`.

Figure 15.6 — Pairwise axis positions across the 14 middle powers  
(colour = \$14 archetype; marker shape = AISI Network membership)**Figure 15:** Pairwise axis positions across the 14 middle powers; colour: archetype; marker shape: AISI Network membership. Source: `outputs/index_baseline.csv`.Figure 16.10 — Sparse-PCA governance sub-axes biplot  
(country positions from per-country sub-axis scores; red arrows = rescaled indicator loadings)**Figure 16:** Sparse-PCA governance sub-axes biplot: country positions with rescaled indicator loadings (red arrows), decomposing the governance axis into institutional-commitment, democratic/multilateral, and cybersecurity sub-axes. Source: `outputs/governance_subaxes.csv`.

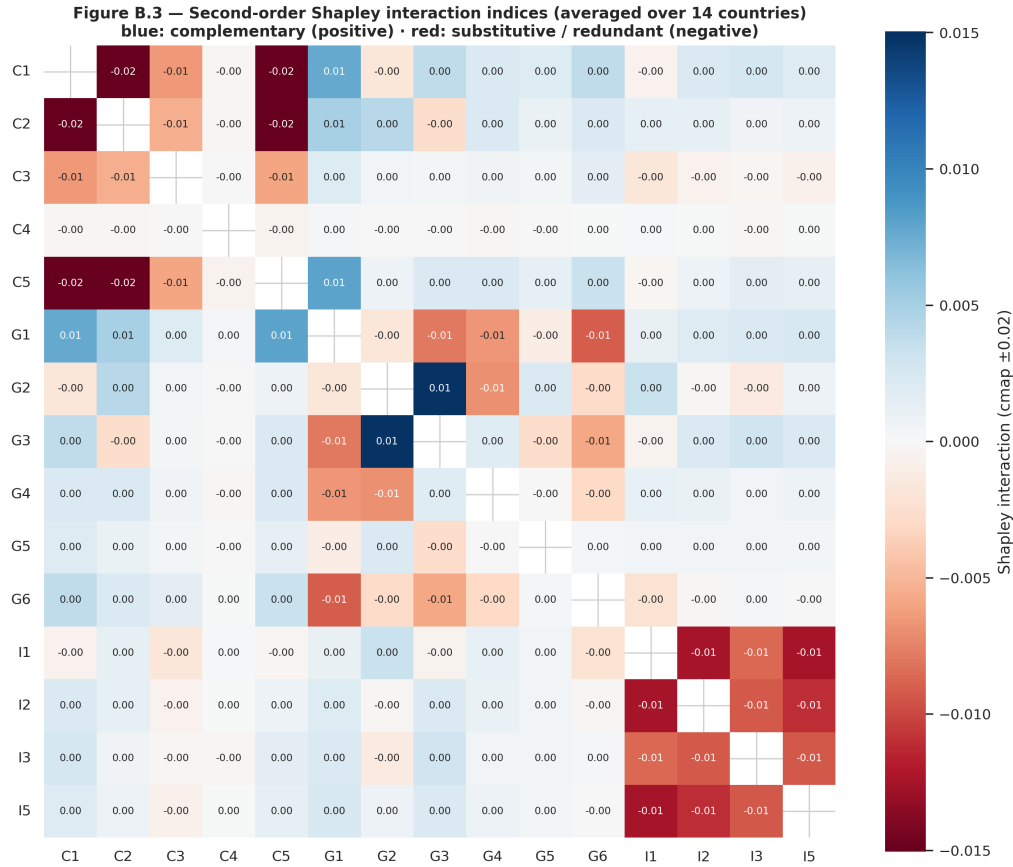


**Figure 17:** Exact Shapley decomposition of the literature composite. Rows: countries by composite rank; columns: indicators; each row sums to the composite minus the universal baseline. Source: `outputs/shap_attributions.csv`.



**Figure 18:** Permutation feature importance for the literature ranking (importance =  $1 - \mathbb{E}[\text{Spearman } \rho]$  between permuted and baseline rankings; 1,000 permutations per indicator), coloured by axis. Source: `outputs/permutation_importance.csv`.





**Figure 19:** Second-order Shapley interaction indices for every indicator pair, averaged over the fourteen countries (all  $2^{15}$  coalitions; [Grabisch and Roubens, 1999](#)). Blue cells are complementary (positive), red cells substitutive/redundant (negative); the diagonal main effect is masked. Substitutive pairs are within-axis; cross-axis pairs, including C5×G1, are weakly complementary. Source: `outputs/shap_interactions.csv`.